How to Know if Something is Random, and if One Randomness Differs from Another

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Abstract

This note sets out Piyush Kumar's original framework for measuring randomness as a deviation from predictability, measured against a fixed reference called a *stencil*, and for comparing two random processes by their relative deviations. It then writes up an extension he modelled: stacking random binary layers, whose limiting behaviour matches the Beer–Lambert law. Sections 1–2 are a faithful transcription of his handwritten notes (photos: `notes-page-1.png`, `notes-page-2.png`). The proofs and simulation in Section 3 were added when the notes were typed up.

1 Questions and definitions

The notes open with two questions: *How do we know something is random?* and *How do we know whether one randomness is different from another?*

Definition 1 (Randomness)**.** Randomness = deviation from predictability.

Definition 2 (Stencil)**.** A *stencil* is a defined predictability: defined feature scores and/or ranges of expected values, against which comparisons are made, acting as an absolute reference.

Both randomness and predictability are calculated against a fixed absolute reference, and in similar features. If features differ, a new temporary feature stencil is created for comparison.

Notation. Let the n-th iteration of a defined stencil be S_n, and let the i-th randomness be r_i.

2 Defined functions

$$\mathrm{Avg}(A, B) = \text{average of } A \text{ and } B, \tag{1}$$

$$\Delta r_i = r_i - S_n, \tag{2}$$

$$\mathrm{Rel}(\Delta, r_i, r_j) = \Delta r_i - \Delta r_j \qquad [\text{where } r_i = r_j \text{ or } r_i \neq r_j]. \tag{3}$$

(In the notes a formula next to $\mathrm{Avg}(A, B)$ is struck out and replaced by the words "Average of A & B". The condition under Rel means it is defined for any pair, whether $r_i = r_j$ or $r_i \neq r_j$; when they are equal, Rel $= 0$.)

Stencil update. The stencil is refined after each iteration:

$$S_{n+1} = \mathrm{Avg}\big(\Delta r_i + S_n,\ S_n\big), \tag{4}$$

which, since $\Delta r_i = r_i - S_n$, simplifies to $S_{n+1} = S_n + \frac{1}{2}\Delta r_i = \frac{1}{2}(r_i + S_n)$: the stencil moves halfway towards each new observation, an exponential moving average with weight $\frac{1}{2}$. If there is more than one randomness, say K of them, then

$$S_{n+1} = \mathrm{Avg}^k\big(\Delta r_k + S_n,\ S_n\big), \qquad k \in [0, K], \tag{5}$$

defining the given randomnesses from r_0 to r_K.

Matrix of stencils. The notes then ask: *what if we make a system of these stencils as a matrix, rather than calculating just one stencil for all features?* This would help manage multiple predictabilities and randomnesses separately, and also let them cross-communicate.

3 Extension: stacked random layers

Piyush extended the framework with the following idea, which he modelled after noticing the pattern in a curtain. It is presented here as an extension Piyush modelled, linking his framework to a known physical law, not as a new theorem.

Take $n \times n$ binary matrices, where 0 is empty space and 1 is filled space. Each layer $L_1, L_2, \ldots$ is filled at random and the layers are stacked parallel to the z–x plane along y. Looking from the front (along y), eventually no position is fully empty; yet if we ignore the count of 1s, every position along y still contains 0s as well. Piyush's conclusion: *randomness extended to infinity can lead to a fixed, defined limit.*

Proposition 1. *Suppose each cell of each layer is filled independently with probability $p \in (0, 1)$. For a fixed position (x, z), after N layers*

$$P(\textit{line of sight clear}) = (1-p)^N \to 0, \qquad P(\textit{line has at least one empty cell}) = 1 - p^N \to 1.$$

Proof. The line at (x, z) meets one cell in each layer, and these N cells are independent. It is clear exactly when all N are empty, which has probability $\prod_{l=1}^{N}(1-p) = (1-p)^N$. It is fully filled when all are filled, probability p^N, so it contains at least one empty cell with probability $1 - p^N$. Since $0 < 1 - p < 1$ and $0 < p < 1$, both $(1-p)^N$ and p^N tend to 0 as $N \to \infty$. □

So both of Piyush's observations hold together in the limit: almost every line is blocked, and almost every line still has gaps. By linearity of expectation, the expected clear fraction of the n^2 lines is also $(1-p)^N$.

Beer–Lambert law. This is the discrete form of the Beer–Lambert law: transmitted intensity decays exponentially with thickness, $T = e^{-\mu N}$ with $\mu = -\ln(1-p)$. Piyush's real-world example is graphene: a single suspended sheet absorbs about $\pi\alpha \approx 2.3\%$ of white light [1], an absorbance of $(2.3 \pm 0.2)\%$ confirmed independently [2]. Ignoring reflection and interlayer effects, N sheets transmit roughly $T \approx 0.977^N$: about 97.7% for one sheet, $\approx 10\%$ for 100 sheets, and essentially 0 for a bulk block of graphite—transparent slice, opaque block.

Simulation. `sim.py` (seed 2026, $n = 200$, up to 60 layers, $p \in \{0.05, 0.10, 0.25\}$) stacks random layers and compares the empirical fractions with theory (Figure 1). For example, at $N = 20$, $p = 0.10$: empirical clear fraction 0.1222 against theory 0.1216.

References

[1] R. R. Nair et al., "Fine structure constant defines visual transparency of graphene", *Science* **320**, 1308 (2008), doi:10.1126/science.1156965.

[2] K. F. Mak et al., "Measurement of the optical conductivity of graphene", *Phys. Rev. Lett.* **101**, 196405 (2008), doi:10.1103/PhysRevLett.101.196405.

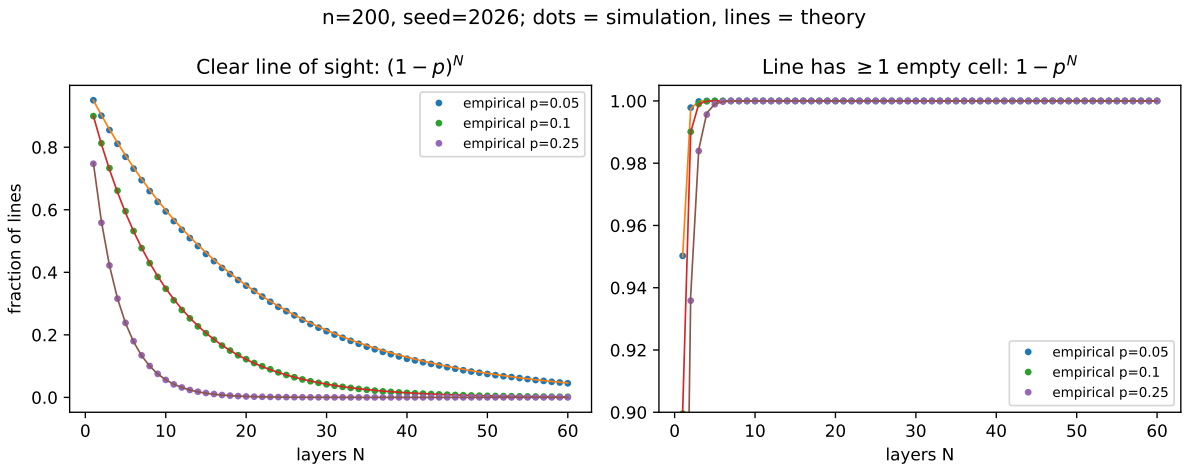
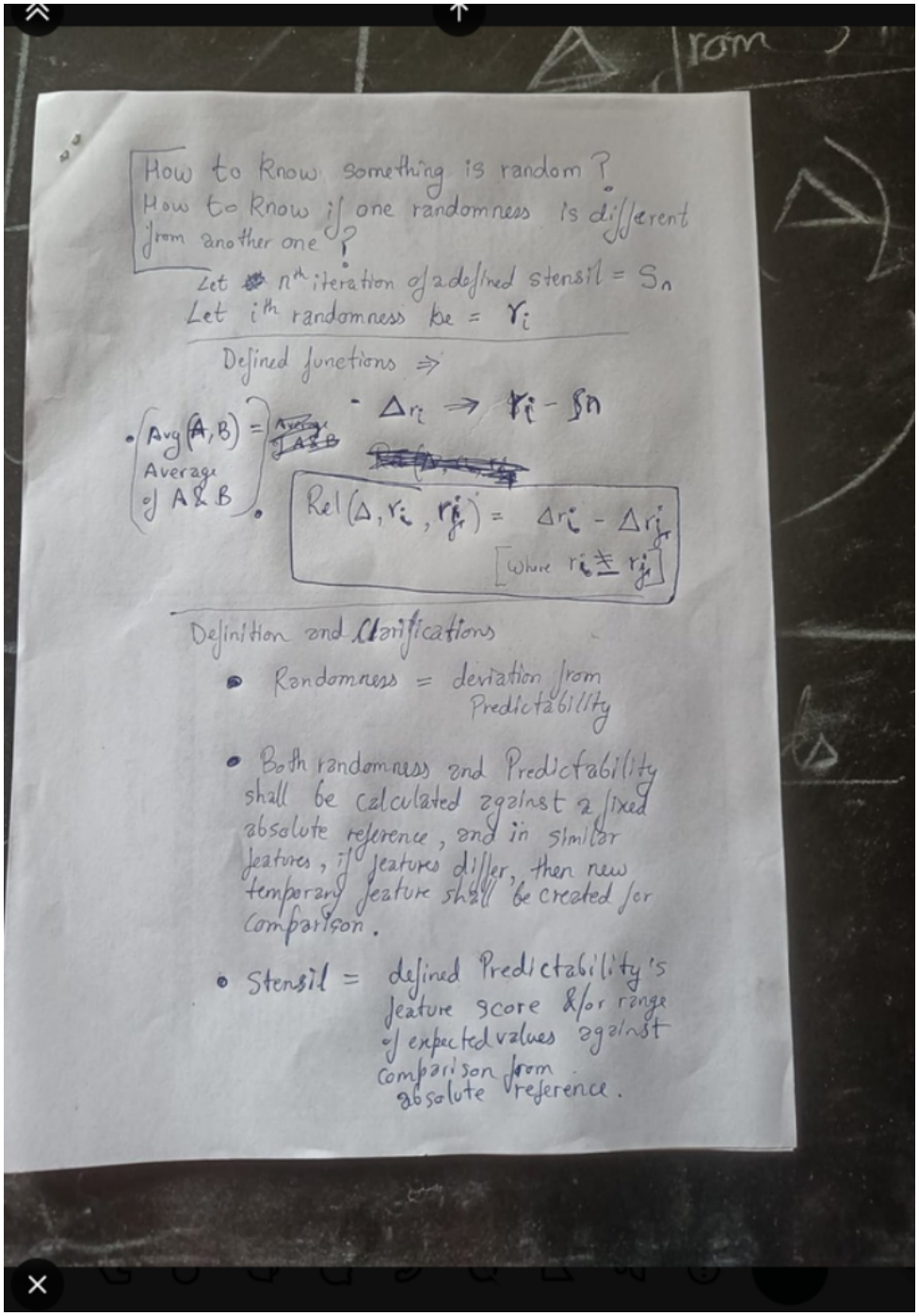
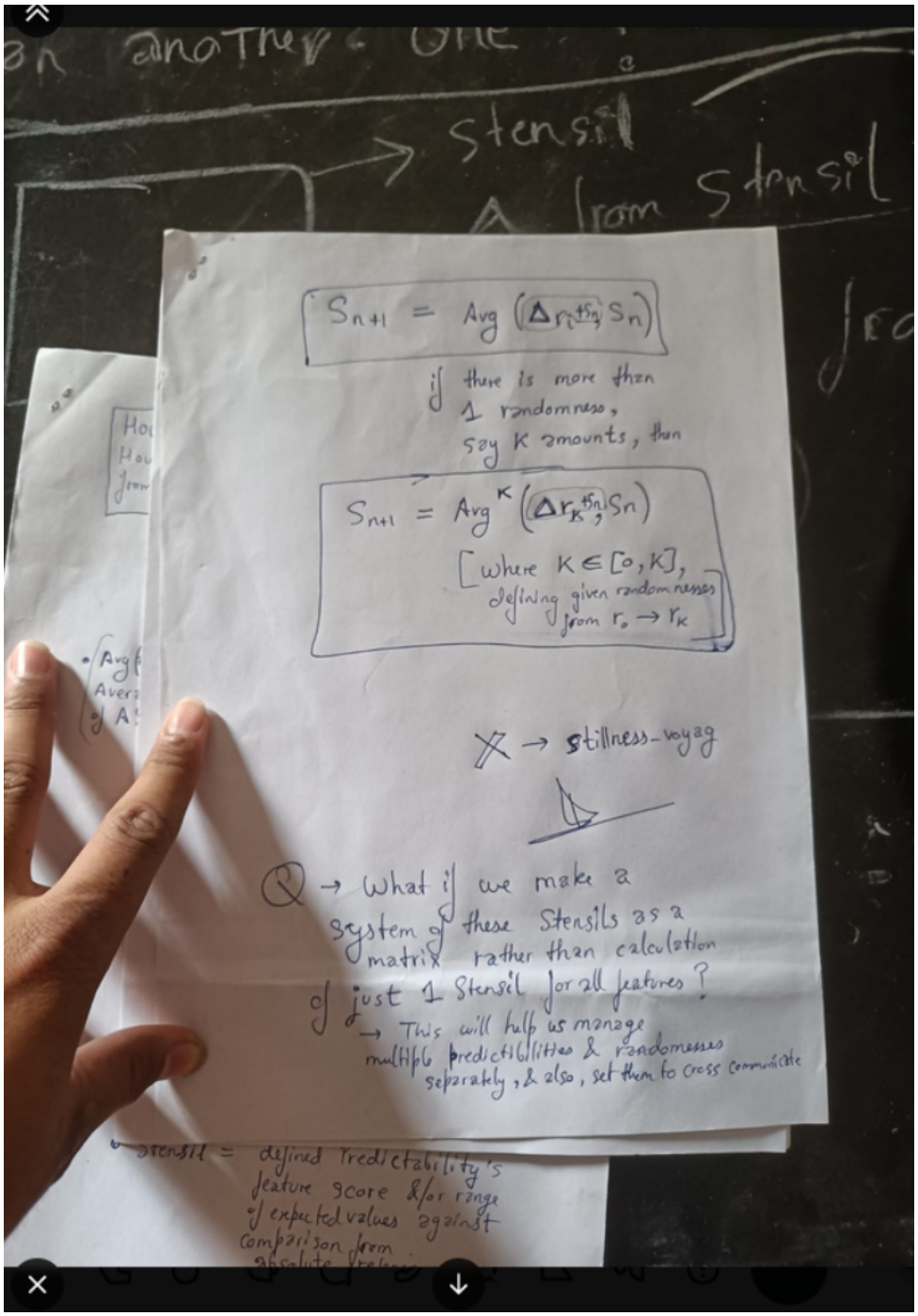


Figure 1: Empirical (dots) versus theoretical (lines) fractions for stacked random layers.

A Original handwritten notes




another one
stensil
from stensil
$S_{n+1} = Avg(\Delta r_{r_1 \to r_2}, S_n)$
if there is more than 1 randomness, say K amounts, than
$S_{n+1} = Avg^{K}(\Delta r_{K}, S_n)$
[where $K \in [0, K]$, defining given randomnesses from $r_0 \to r_K$]
stillness-voyag
Q → What if we make a system of these Stensils as a matrix rather than calculation of just 1 Stensil for all features?
→ This will help us manage multiple predictibilities & randomesses separately, & also, set them to cross communicate
Stensil = defined predictability's feature score &/or range of expected values against comparison from